

Specification: A Continuous-Quote RFQ Market for Perpetual Options on a Burr-II Kernel, with its Formal (Lean 4) Backing

Temporal Exchange working notes*

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Abstract

We specify a request-for-quote (RFQ) market-maker console for perpetual options in which a maker's quote at *every* strike is a single closed-form curve: four parameters ($\bar{S}, a, \gamma, \kappa$) of a Burr-II (Burr Type XII in the moneyness modulus) wing kernel, plus a half-spread h . There is no order ladder: any strike is priced on demand, requoting is four parameters rather than N orders, and the two curve sets (bid and ask) make the spread an irreversible revenue band. Makers differ only in depth and spread while sharing the price *level* — a design forced by a proved theorem (heterogeneous per-maker steepness generates a strict volatility smile that no single power-law lens can represent). The tradeable book is the envelope of maker curves; RFQ fills walk makers cheapest-first, each paid at its own posted spread; settlement nets in perpetual units and crosses one common units→cash doorway, which is exactly the arbitrage-free class. We give the closed form, the quoting economics (vol-indexed spreads against gamma bleed), the aggregation and fill rules, the user-interface contract, worked numbers re-derived from the shipped code, and the complete formal corpus backing the design: five Lean 4 packages, 130 theorems, all kernel-checked locally with axiom set {propext, Classical.choice, Quot.sound}. Open gaps are stated rather than hidden.

1 Scope and provenance

Object specified. `app/index.html` (a single 562-line file; Burr-II math in JavaScript including a continued-fraction regularized incomplete beta), `app/server.js`, `app/railway.json`. Build 11 = commit `ebd774e` (2026-08-14); deployed at <https://web-production-cc265.up.railway.app>.

Status. The console is a *brainstorm simulation* (its own footer says so): it is not the production engine (`engine/builds/HEAD_temporal_mvp_v28_lens.html`, which is untouched by all of this), and it uses demo constants — spot $S = \$65,695.5$, three synthetic competitor makers, demo portfolio positions.

Verification posture. Every number in §6 was re-derived on 2026-08-18 by executing the shipped code's own mathematics in Node.js, not copied from build logs. The Lean corpus of §7 was proved by the external prover Aristotle (Harmonic) or written in-repo, then *locally*

*Specification of the market-maker console `app/index.html`, builds 1–11 (2026-08-14), repository `Perp-Options-AMM`. Prepared 2026-08-18. This document specifies a *brainstorm simulation*, not the production engine; see §1.

kernel-verified in our own Lean 4.28.0 + Mathlib kernel; caveat stated, not hidden: Mathlib's `.olean`s came from the Mathlib CI cache, the universal practice, not a from-scratch rebuild. The JavaScript $\leftrightarrow$ Lean correspondence is statement-by-statement mirroring plus numeric spot checks; the transcription itself is asserted, not proved.

2 The pricing kernel

Let S be spot, K a strike, and $k = K/S - 1$ the linear moneyness (calls are out of the money for $k \geq 0$, puts for $k \leq 0$; the zero-strike edge is $k = -1$). Fix parameters $\bar{S} > 0$ (mean strike scale), $a > 0$ (Burr shape), $\gamma > 0$ (tail weight), $\kappa \in (-1, 1)$ (skew). Write $B = \Gamma(\frac{1}{a})\Gamma(\frac{\gamma}{a})/\Gamma(\frac{1}{a} + \frac{\gamma}{a}) = B(\frac{1}{a}, \frac{\gamma}{a})$ and let $I_u(\cdot, \cdot)$ be the regularized incomplete beta function.

Wing scales and kernel.

$$s_R = \bar{S}(1 + \kappa), \quad s_L = \bar{S}(1 - \kappa), \quad \Delta_s(v) = \left(1 + (v/s)^a\right)^{-\frac{\gamma+1}{a}}. \quad (1)$$

Wing value (tail integral), pedestal, masses.

$$T_s(m) = \frac{s}{a} B \left(1 - I_u\left(\frac{1}{a}, \frac{\gamma}{a}\right)\right), \quad u = \frac{m^a}{m^a + s^a}, \quad G_1 = \Delta_{s_L}(1), \quad I_1 = T_{s_L}(1), \quad (2)$$

$$W_R = \frac{s_R B}{a}, \quad W_L = \frac{s_L B/a - I_1 - G_1}{1 - G_1}. \quad (3)$$

The peg (derived, not chosen).

$$q_R = \frac{W_L}{W_R + W_L}, \quad q_L = 1 - q_R \quad (4)$$

is the *unique* split making the two wings meet at the money (Lean: `atm_wings_meet`); at-the-money put-call parity *is* the peg.

Wings and option values.

$$A_R(k) = q_R T_{s_R}(|k|), \quad (5)$$

$$A_L(k) = \begin{cases} \frac{q_L}{1 - G_1} \left(T_{s_L}(|k|) - I_1 - G_1(1 - |k|)\right), & |k| < 1, \\ 0, & |k| \geq 1, \end{cases} \quad (6)$$

$$\text{CALL}(k) = \begin{cases} A_R(k), & k \geq 0 \\ -k + A_L(k), & k < 0 \end{cases} \quad \text{PUT}(k) = \begin{cases} A_L(k), & k \leq 0 \\ k + A_R(k), & k > 0. \end{cases} \quad (7)$$

Out of the money the value is the wing; in the money it is intrinsic plus the *mirror* wing, so

$$\boxed{\text{CALL}(k) - \text{PUT}(k) = -k \quad \text{at every strike}} \quad (\text{parity})$$

(Lean: `burr2_parity`; away from $k = 0$ it is hypothesis-free branch bookkeeping, `burr2_parity_of_ne`). The left wing's closed form already vanishes at the zero-strike edge (`AL_glue`), giving the exact anchors $\text{CALL}(-1) = 1$ (the whole coin) and $\text{PUT}(-1) = 0$. The at-the-money mark is

ATM = $q_R W_R$. Sign and order — all values nonnegative, CALL antitone and Lipschitz, PUT monotone — hold conditionally on a two-sided Riemann-sandwich interface for the incomplete beta (**TailRep**, proved non-vacuous by an explicit witness); see §8.

3 Quoting: two curve sets and vol-indexed spreads

The curve is the quote. A maker posts $(\bar{S}, a, \gamma, \kappa)$ and a half-spread h (bps). At any strike, on demand,

$$\text{ask}(k) = V(k) \left(1 + \frac{h}{10^4}\right), \quad \text{bid}(k) = V(k) \left(1 - \frac{h}{10^4}\right), \quad (8)$$

with $V \in \{\text{CALL}, \text{PUT}\}$. There is no ladder to post, cancel, or replace; requoting is four parameters, not N orders. The band between the two curve sets is the maker’s revenue and is *irreversible* (unlike constant-product-AMM slippage, which a round trip refunds). Execution is priced on the *continuous (integral) convention*: impact is a price path, not maker revenue — the endpoint-price shortcut that fabricated “slippage revenue” was identified and removed before this app was built (operator rulings, transcript entries 539–540).

Gamma bleed and the fair spread. With cash value $C_{\text{cash}}(S, K) = \text{CALL}(K/S - 1) \cdot S$, the console measures a per-dollar-notional cash curvature

$$\tilde{G} = \text{mean}_{K/S \in \{0.85, 0.95, 1, 1.05, 1.15\}} \left| S \partial_S^2 C_{\text{cash}}(S, K) \right|, \quad \text{bleed} = \frac{1}{2} \tilde{G} R V^2 \text{ [notional/yr]}, \quad (9)$$

and the fair half-spread and quote

$$h_{\text{fair}} = \frac{\text{bleed}}{\text{turnover} \cdot 365} \times 10^4 \text{ bps}, \quad h_{\text{eff}} = \begin{cases} h_{\text{fair}} \cdot (1 + 25\%) & \text{vol-indexed mode,} \\ \text{fee} \cdot 10^4 / 2 & \text{fixed-spread mode.} \end{cases} \quad (10)$$

Fees, bleed and returns per year on quoted notional $N_{\$}$ (margin $\times$ 10 pool leverage, both wings) are

$$\text{fees} = \frac{h_{\text{eff}}}{10^4} \text{turn} \cdot 365 \cdot N_{\$}, \quad \text{bleed}_{\$} = \text{bleed} \cdot N_{\$}, \quad \text{APR}_{\text{base}} = \frac{\text{fees} - \text{bleed}_{\$}}{N_{\$}}, \quad \text{APR}_{\text{lev}} = \frac{\text{fees} - \text{bleed}_{\$}}{\text{equity}}. \quad (11)$$

Capacity is a *continuous* density $w(k) \propto e^{-8\lambda k^2}$ normalized to the wing notional — the ribbon in the chart, not rungs. The economics layer is a model with no Lean backing (§8).

4 Aggregation, the RFQ, and settlement

Common level, private transport. All makers share the level parameters $(\bar{S}, a, \gamma, \kappa)$; each maker owns only its depth λ_i and half-spread h_i . This is not a stylistic choice: **mixture_not_single_lens** (§7) proves that mixing distinct per-maker steepnesses produces a strictly log-convex (smiled) aggregate that *no* single power-law lens represents, and build 11’s own test reproduced the failure empirically — per-maker levels crossed the book (best bid 0.15482 > best ask 0.12015, spread −1,261 bps). The shipped design is “transport, not level,” with an honest **CROSSED** — arb detector that cannot fire under a common level and positive spreads.

Depth aggregation (harmonic).

$$\frac{1}{\lambda_{\text{agg}}} = \sum_i \frac{1}{\lambda_i}, \quad \text{share}_i = \frac{1/\lambda_i}{\sum_j 1/\lambda_j} \quad (12)$$

(Lean: `harmonic_law`, `shares_sum_one`, `share_eq_agg_ratio`; the parallel-addition law itself is *earned* in the base layer from walking a stacked ladder to a common marginal price, `walk_equiv/walk_cost_equiv`, interior case).

Envelope book and RFQ fill.

$$\text{ask}(k) = \min_i V_i(k) \left(1 + \frac{h_i}{10^4}\right), \quad \text{bid}(k) = \max_i V_i(k) \left(1 - \frac{h_i}{10^4}\right), \quad (13)$$

so the aggregate is always at least as tight as any single maker (the “improvement vs. single maker” panel quantifies what competition buys). An RFQ of size Q walks makers cheapest-first, fills each up to $\text{share}_i \times \text{pool}$, and pays each maker *at its own posted spread* (pay-your-own-quote, not uniform-price; Lean: `apportionment_conserve`).

Settlement doorway. Legs net in perpetual *units*; cash appears only through one common conversion at the exit. This is forced, not convenient: `doorway_arbfree_iff_common` proves a book netting to zero units pays zero cash *iff* the units→cash factor is common across strikes, and any per-strike factor admits unbounded arbitrage (witness: long $1/V_i$ at strike i , short $1/V_j$ at strike j — zero net units, cash $F_i - F_j$, scalable). `exit_timing_irrelevant`: a netted-flat book settles to zero for every closing equity and leverage.

Trade bands and portfolio. A band sells one wing and buys the other with the proceeds (per-leg fee and slippage); *only κ moves* — level and shape stay put — and every strike re-prices at once. The portfolio view keeps perps and perp-options in one account equity (account-level liquidation, 50× cap with headroom), carves option-backing notional out of the perps tab, and reads net option delta back to a perp hedge with zero residual. The κ dynamics and the exposure readback have *no* Lean backing (§8).

5 User-interface contract

view	contract
EARN (maker)	Curve parameters $\bar{S} \in [.05, 1]$, $a \in [.2, 3]$, $\gamma \in [.3, 4]$, $\kappa \in [-.9, .9]$, $\lambda \in [.01, 1]$, fee $\in [0, .2]$, depth $\in [1, 80]$ BTC/1%-strike; margin. Chart: call/put ask solid, bid dashed, both first-class; spread drawn as a filled band; “spread view $\times N$ ” magnifier (default $\times 25$) with an honest caption — at ~ 19 bps the true gap would overlap on screen; mid demoted to faint reference; capacity-density ribbon. <i>Hover quoting</i> : a crosshair tracks any strike; readout = strike, call bid/ask, put bid/ask, half-spread, $C - P$ vs $-k$ parity, spread capture on size. No standalone quote box (the chart <i>is</i> every strike). Market sliders: realised vol, turnover/day, RFQ size. Risk: curvature, break-even half-spread and turnover, parity check to 12 dp. Vol-indexed toggle.
TRANSACT (taker)	Buy/sell, any strike $k \in [-60\%, +60\%]$, size. Envelope best-ask solid / best-bid dashed with individual makers faint behind; order ticket prices off the envelope and names the filling maker; fill table walks makers cheapest-first; book state flag clean / CROSSED — arb.
TRADE BANDS	Sell-strike, buy-strike, size; both leg prices, proceeds, units bought; state move $\kappa \rightarrow \kappa + \Delta\kappa$ with $\bar{S}, a, \gamma$ explicitly unchanged; curve drawn before/after.
PORTFOLIO	Positions with marks and deltas; one account equity, leverage vs the $50\times$ cap with headroom and SAFE/LIQUIDATED ; carve panel; hedge readback (net option $\Delta \rightarrow$ required perp hedge $\rightarrow$ residual 0).

6 Worked numbers (re-derived from the shipped code)

Defaults: $\bar{S} = .60$, $a = 1.2705$, $\gamma = 1.8413$, $\kappa = 0$, $\lambda = .10$, fee = .02, margin 15 BTC, $S = \$65,695.5$, RV 60%, turnover $0.30\times/\text{day}$, pool 200 BTC, RFQ 5 BTC. All values recomputed 2026-08-18 by executing the app’s own math in Node.

quantity	value	note
ATM mark	0.178888	$q_R W_R$
CALL/PUT at $k=+12\%$	0.135450 / 0.255450	Transact default strike
parity $ C - P + k $, $k=0.2$	0	exact in floating point off-ATM
parity $C - P$ at $k=0$	2.44×10^{-13}	the peg, numerically (continued-fraction I)
CALL(-1) / PUT(-1)	1 / 0	zero-strike anchor, exact
curvature $\tilde{G}$	0.9065	five-point mean
bleed	0.16317/yr	fraction of notional
$h_{\text{fair}} / h_{\text{eff}}$	14.90 / 18.63 bps	fair; vol-indexed +25%
Σ notional	300 BTC = \$19,708,650	margin $\times 10$, both wings
fees / bleed per yr	\$4,019,813 / \$3,215,850	
base / levered APR	4.1% / 81.6%	levered on 15 BTC equity
break-even turnover	$0.24\times/\text{day}$	vs $0.30\times$ assumed
envelope at $k=12\%$	ask 0.13567 / bid 0.13523	book spread 32.8 bps (best-maker half-spr
depth shares	47.6/19.0/9.5/23.8 %	YOU/Kappa/Delta/Sigma, harmonic
band (sell +10% 5 BTC, buy -10%)	proceeds 0.6840 BTC, buys 5.2346 u	$\Delta\kappa = +0.00574$; $\bar{S}, a, \gamma$ unchanged

Vol sweep (levered APR, current build-11 code).

realised vol	h_{fair} (bps)	vol-indexed	fixed (fee = 2%)
20%	1.7	+9.1%	+2153.7%
60%	14.9	+81.6%	+1863.7%
120%	59.6	+326.3%	+884.6%

Finding. At the shipped default fee = 0.02 (= 100 bps half-spread) the fixed mode never inverts within the RV slider range [10%, 140%] — inversion needs $RV > 155\%$ at fee 2%, or fee $\lesssim 1\%$ — so the in-app fixed-mode caption (“watch the APR invert”) only demonstrates at lower fee settings. The build-5 headline (vol-indexed +3.1/ + 27.7/ + 110.8% vs fixed +24.5/ − 74.0/ − 406.4% APR at RV 20/60/120%) is *historical*: build-5 code and calibration, not reproducible on build-11 defaults.

7 The formal corpus

Five Lean 4 packages, 130 theorems. Uniformly: no `sorry/admit/axiom/native_decide/opaque/unsafe`; every theorem’s `#print axioms = {propxt, Classical.choice, Quot.sound}`; statements byte-identical to what was submitted (prover-side edits are proof bodies and private helpers only); all locally kernel-verified in Lean 4.28.0 (Mathlib .oleans from the CI cache — stated, not hidden). Label-flip from *trusted-from-prover* to *verified* is on record as a recommendation; the call is the operator’s.

7.1 BURR2_CORE — 42 theorems (Aristotle run 26958f2b)

Theorems about the *exact* production formulas of §2, conditional on nothing but four interface facts about the incomplete beta (carried as a structure field, not an axiom). §1 sheet correspondence and positivity (`uArg_eq_sheet`, `tail_zero`, `G1_pos`, `G1_lt_one`, ...); §2 parity (`burr2_parity_of_ne` hypothesis-free off-ATM; `atm_wings_meet`: the peg is *derived* as the unique wing-meeting split; `burr2_parity`); §3 zero-strike anchor (`AL_glue`, `CALL(−1) = 1`, `PUT(−1) = 0`); §4 sign and order (19 theorems: kernel in $(0, 1]$ decreasing, wings nonnegative, `CALL_antitone`, `CALL_lipschitz`, `PUT_monotone` — conditional on `TailRep`); §5 non-vacuity witness; §6 apportionment (`harmonic_law`, `shares_sum_one`, `share_eq_agg_ratio`, `share_nonneg`, `apportionment_conserves`). Two substantive corrections the proofs forced on the brief: at-the-money parity *is* the peg (the 1.39×10^{-17} and 2.44×10^{-13} residuals are one phenomenon, not two), and $q_R = W_L/(W_R + W_L)$ is derived, not chosen.

7.2 BURR2_MIXTURE — 13 theorems (run e95f6377)

Burr-II is *not* closed under maker mixture (`burr2_not_closed_under_mixture`, and the §5 stretch version for scale mixing via a two-atom moment/Cauchy–Schwarz argument), *but* the exact identity

$$\frac{1}{2}(\Delta_{\gamma_1} + \Delta_{\gamma_2}) = \Delta_{\bar{\gamma}} \cdot \cosh(\delta L_c) \quad (14)$$

bounds the mixture as the same shape times $1 + O(\delta^2)$ — *second* order in the parameter gap, versus the single-lens wall of §4 which is first-order structural. The kernel change is what buys that mildness.

7.3 LOOP_LINK_A (pricing) — 9 theorems (project 20fa5993, task b1e0c962)

The pricing↔aggregation bridge. `engine_call_midconvex` ($g \geq 1$, via a global supporting line) discharges the standing MidConvex hypothesis for the v28 engine curve; `atm_kink_bound` in the stronger form $(g+1)\left(\frac{g+1}{g}\right)^g \geq 2g+1$ for all $g > 0$; `engine_book_arb_free` / `engine_book_parity`: a book of makers each with its own exponent g_i is butterfly-arbitrage-free and parity-anchored. And the load-bearing negative half: `engine_log_affine` + `mixture_strict_log_convex` $\implies$ `mixture_not_single_lens` — *heterogeneous maker steepness generates a strict smile that no single power-law lens can represent*. This proved obstruction is why the console’s makers share the level (§4).

7.4 LOOP_LINK_B (settlement) — 12 theorems (project ce3c2190, task aa5f90f8)

Netting in units is linear and well-defined (`units_add`, `units_smul`, `cashOne_add`); the common doorway neither creates value nor flips signs (`cashOne_zero_iff`, `cashOne_pos_iff`); it cannot be timed (`exit_timing_irrelevant`); and the pair `common_doorway_necessary` / `per_strike_doorway_unbounded_a` yields `doorway_arbfree_iff_common`: one common units→cash factor is *exactly* the arbitrage-free class. “Dollars enter only at the very end” is a theorem.

7.5 sims/v3-maps-lean (base layer) — 54 theorems

The book/aggregation substrate (11 BOOK + 25 MAP + 18 BASIS): `agg_parity` (weights summing to one), `agg_midconvex` (nonnegative weights; fixed-across-strikes enforced structurally), `book_arb_free` (butterfly leg), `level_convex_iff_zero_spread_arbfree` (a genuine iff, the strongest statement in the file), `share_strike_invariant`, `beta_transport_parallel` with the parallel-addition law earned by `walk_equiv/walk_cost_equiv` (interior case), and the exact vega dichotomy `shares_fixed_of_common_transport` / `common_transport_is_necessary`.

8 Open gaps (stated, not hidden)

1. **The Burr-II MidConvex bridge is not proved.** The book-layer arbitrage-freeness chain attaches to the v28 engine power-law curve (`engine_call_midconvex`) and only *hypothesis-conditionally* to the Burr-II curve this console actually quotes. Named the next Lean obligation.
2. BURR2_CORE §4 sign/order results are conditional on `TailRep`; the non-vacuity witness is a toy ($a = \gamma = 1$, $I = \text{id}$), not the real incomplete beta.
3. The BURR2_MIXTURE bound is on the midpoint member, near-the-money in scope (residuals of 0.02–0.12% are near-money; 745% excess at γ 0.4 vs 3.0); joint non-closure over all four parameters is unproved.
4. κ (skew) dynamics — the Trade Bands mechanic — are entirely unformalised.
5. **Mislabel risk (base layer):** the exposure readback (`Exposure`) is cited in v3-maps-lean’s README and the loop map but is *absent* from the Lean files (re-verified 2026-08-18); the Portfolio hedge-readback panel has zero Lean backing.

6. “Arb-free” means *butterfly* only: no unconditional vertical leg, no price bounds ($C \geq$ intrinsic, $C \leq S$); and there is no individual-rationality/adverse-selection theorem for makers under the mixed book (the refraction finding).
7. The economics layer (bleed, h_{fair} , APR, viability) is a model with demo constants and no Lean; the fixed-mode inversion is unreachable at the default fee (see §6). Realtime maker tuning converts a passive-LP product into an active-MM venue — adverse selection, correlated withdrawal, and herding are new, unmodelled risks (operator transcript entry 541).
8. The console is not wired to the production engine; nothing here changes **engine/** or its gates.

A Build history (all 2026-08-14)

Operator-direction fragments below are quoted from the build commit messages; the session’s verbatim transcript is missing (corrigendum filed 2026-08-18, `history/operator/CORRIGENDUM_2026-08-18_appbuild`).

build	commit	change
1	75abb52	Burr-II RFQ book, single file, Railway-ready; parity exact headless.
2	269a6a3	Deployed to Railway (PORT fix); BURR2_CORE + BURR2_MIXTURE Lean landed (55 theorems).
3–4	0e2b10a, 8dfd02e	Rebuild on the reference design system; gamma-bleed basis fixed to notional.
6	caf6c71	Rung ladder stripped — continuous quoting (operator: “that was the OB”; take the UX elements, not the orderbook mechanics).
8	07370be	Bid and ask as first-class separate curves; filled spread band; $\times N$ magnifier with honest caption; build stamp.
9	7650a2a	Sampled quote sheet replaced by an any-strike quoter.
10	7fd1325	Quote box merged into the chart — hover to quote (operator: “why u need box separately”).
11	ebd774e	All views activated; aggregate book on Transact; <i>crossed-book bug found and fixed</i> — per-maker levels crossed the book; makers now share the level and differ in depth/spread only; CROSSED — arb detector added.

Twin document: `SPEC_RFQ_MM_CONSOLE.xlsx` in the same directory carries this specification in spreadsheet form, including the theorem-level listing of all 64 RFQ-specific theorems.